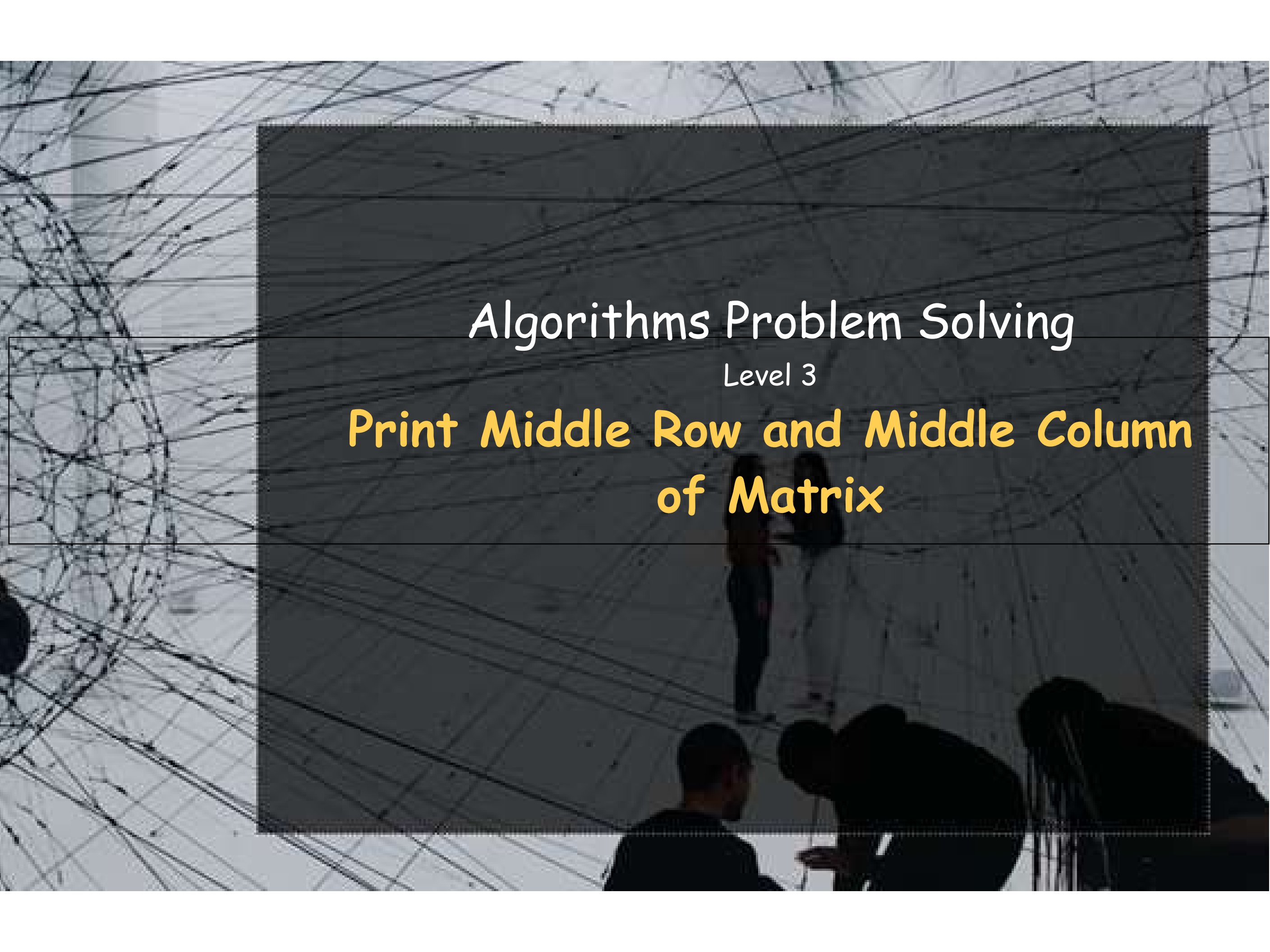


Algorithms Problem Solving

Level 3

Print Middle Row and
Middle Column of Matrix



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Definition

 Middle Row & Column

 Program generates a 3×3 matrix of random numbers and extracts/displays its middle row and middle column.

Explanation

Index Calculation

 For odd dimension N , middle index is $N / 2$ (e.g., index 1 for $N=3$).

Procedures

1. Populate 3×3 matrix with random values (1 to 10).
2. Print full matrix.
3. Extract and print row at index `rows / 2` with `PrintMiddleRow()`.
4. Extract and print column at index `cols / 2` with `PrintMiddleColumn()`.

💡 Example

Matrix1:

01 10 07

10 03 02

06 03 09

Middle Row:

10 03 02

➡ Result

Middle Column:

10 03 03

Note

- ◆ Adjust index calculation logic if handling even-sized matrices.

⊘ Warning

- ◆ Matrix dimensions must be odd to have a single exact center row or column.
- ◆ Using hardcoded index `1` instead of `rows / 2` breaks flexibility for larger matrices.
- ◆ Out-of-bounds errors occur if row/col calculations exceed matrix dimensions.



Comparison

▨ `PrintMiddleRow()`

- ✓ Accesses matrix elements along a fixed row index (`matrix[middleRow][j]`).
- ✓ Traverses horizontally across columns.
- ✓ Outputs elements in a single horizontal line.

▤ `PrintMiddleColumn()`

- ✓ Accesses matrix elements along a fixed column index (`matrix[i][middleCol]`).
- ✓ Traverses vertically across rows.
- ✓ Outputs elements vertically or separated by tabs.

✦✦ Summary

- ✓ Calculates the middle coordinate index for odd-dimension matrices using integer division ($\text{dimension} / 2$).
- ✓ Isolates and outputs the horizontal vector representing the center row.
- ✓ Extracts and outputs the vertical vector representing the center column independently.

Thank you 😊